

# VALVE POWER BOARD

## RUBY - REV A



RESPONSIBLE ENGINEER:  
NICHOLAS ALVES

REVIEWERS:

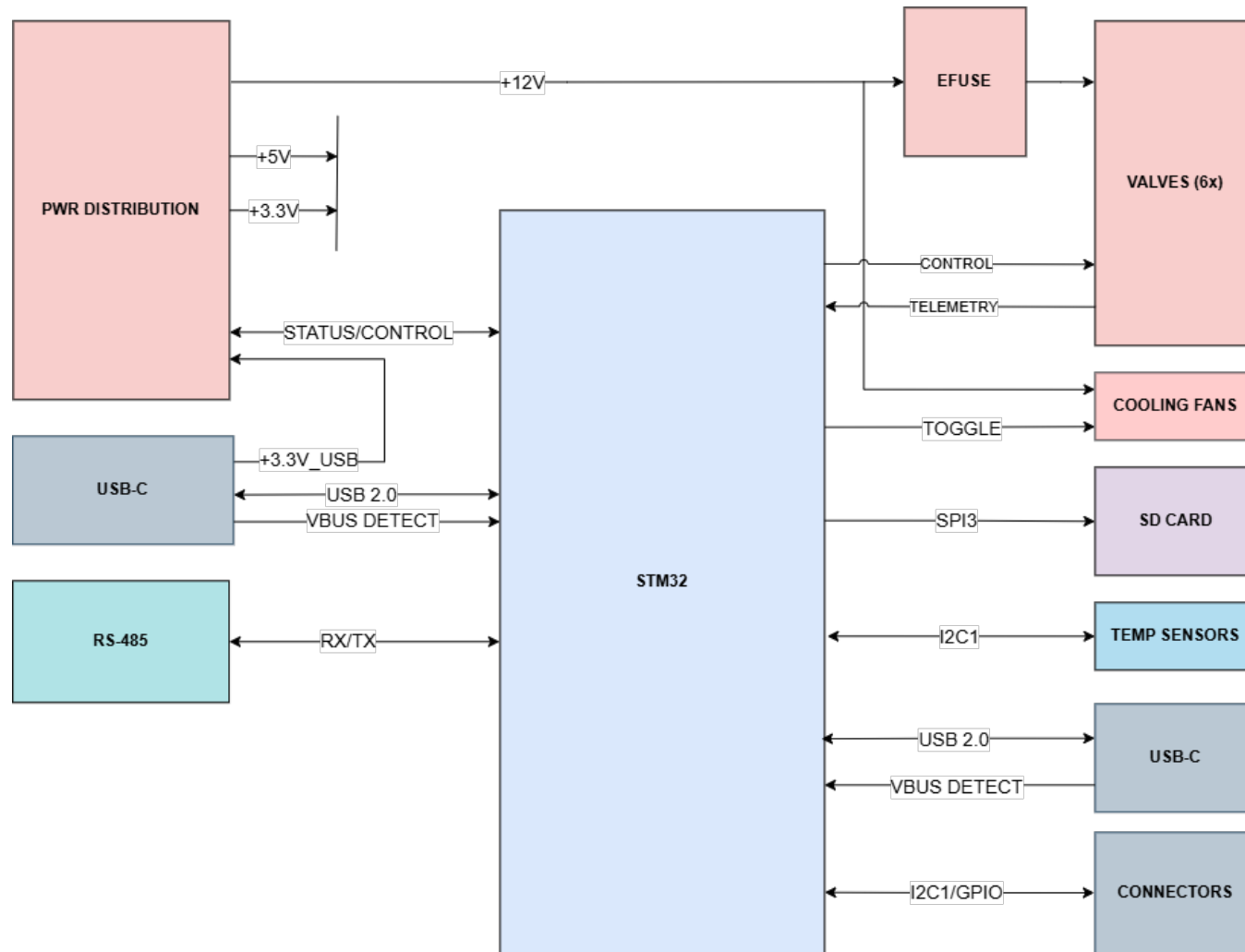
**SCRT**  
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/  
File: top-level.kicad\_sch

**Title: TITLE PAGE**

Size: A4 Date: 2026-08-27  
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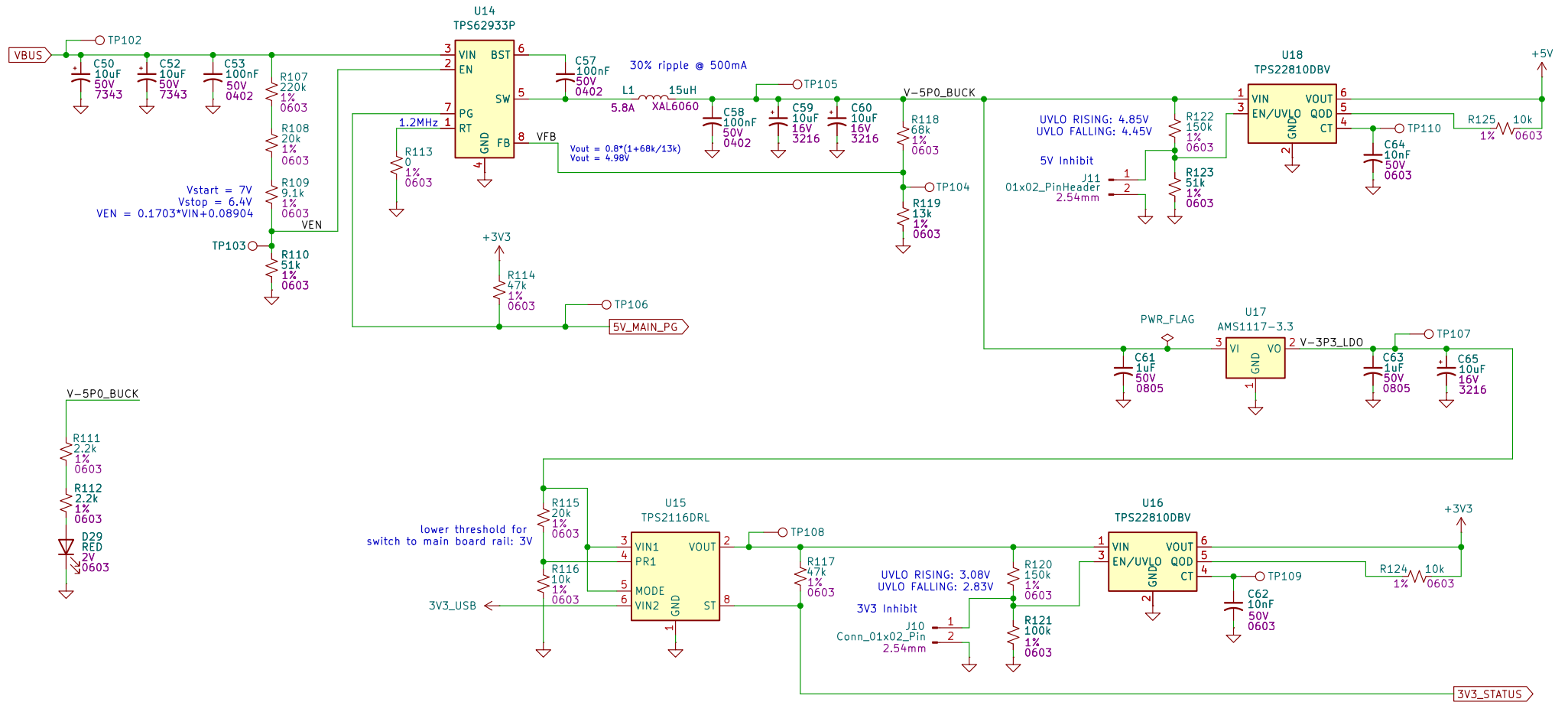
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File: block\_diagrams.kicad\_sch

**Title: BLOCK DIAGRAM**

Size: A4 Date: 2026-08-27  
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# SYSTEM POWER



Sheet: /System Power/  
File: system\_power.kicad\_sch

**Title: SYSTEM POWER**

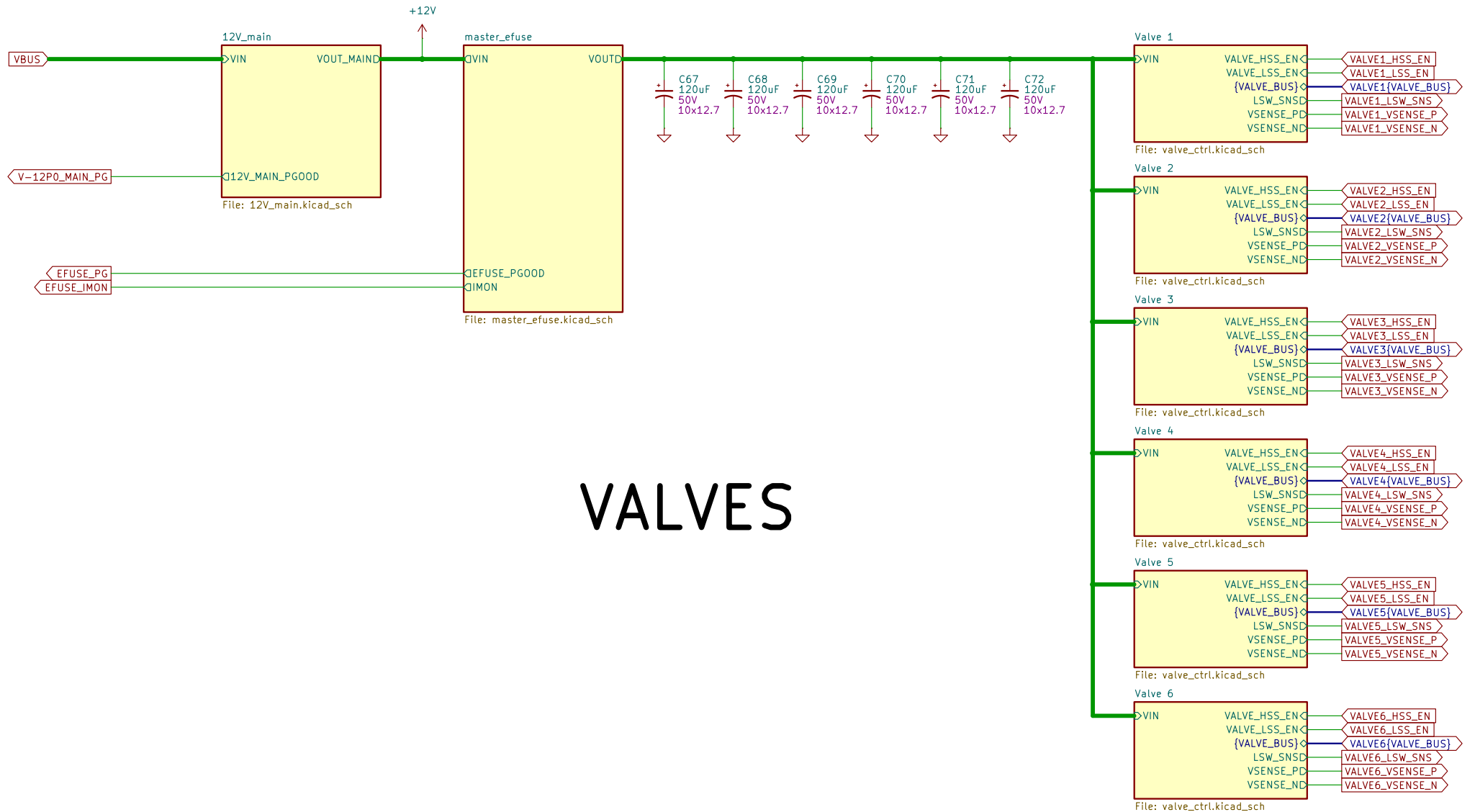
Size: A4 Date: 2026-08-27

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# VALVES



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**Title: VALVES**

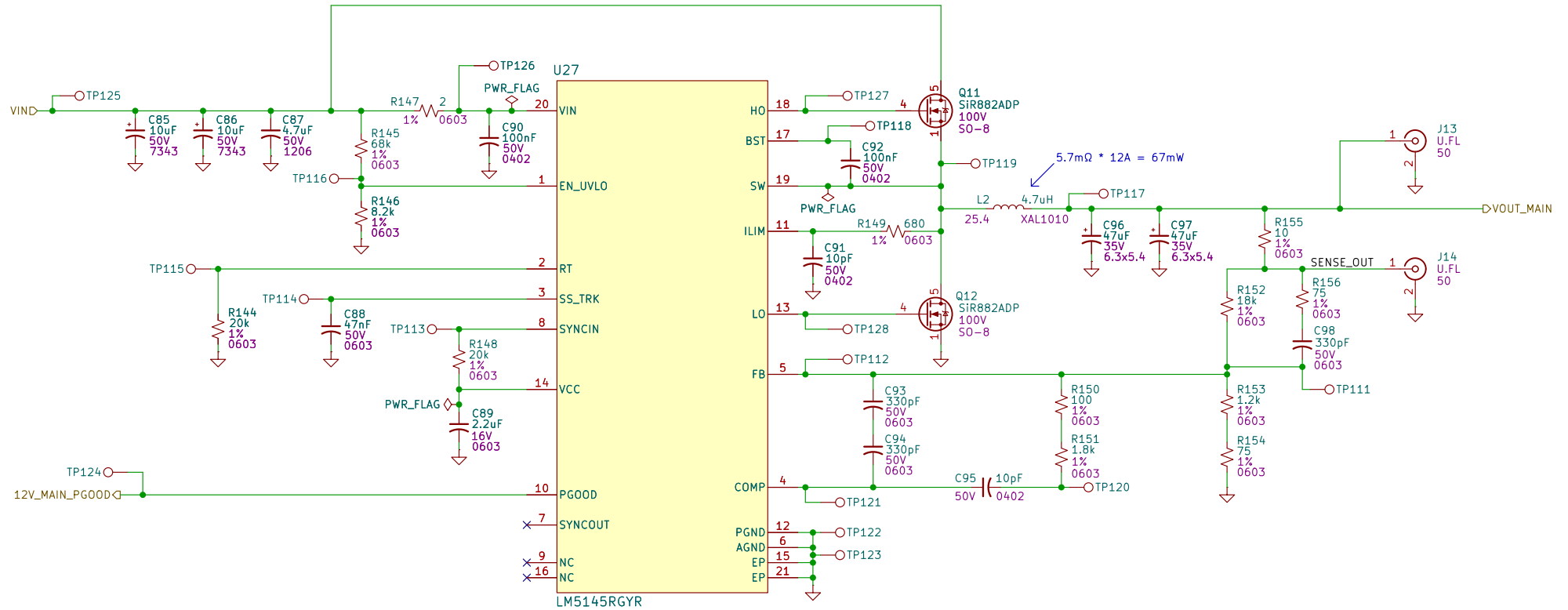
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# 12V MAIN



Sheet: /Valves/12V\_main/  
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**Title: 12V MAIN**

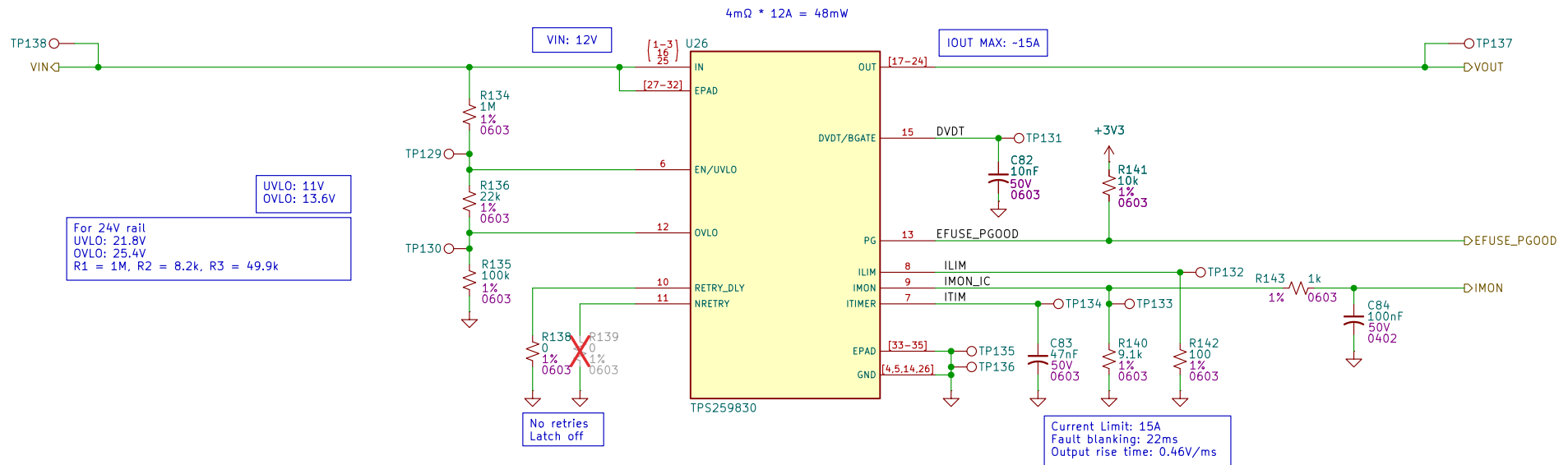
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# EFUSE



Sheet: /Valves/master\_efuse/  
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Title: EFUSE

Size: A4 Date: 2026-08-27

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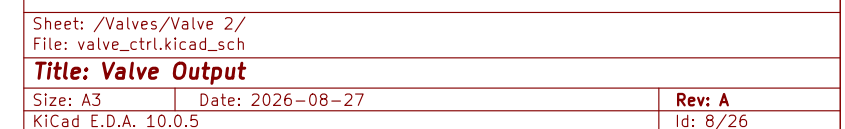
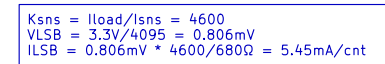
Id: 6/26

## VALVE OUTPUT POWER

[illegible]

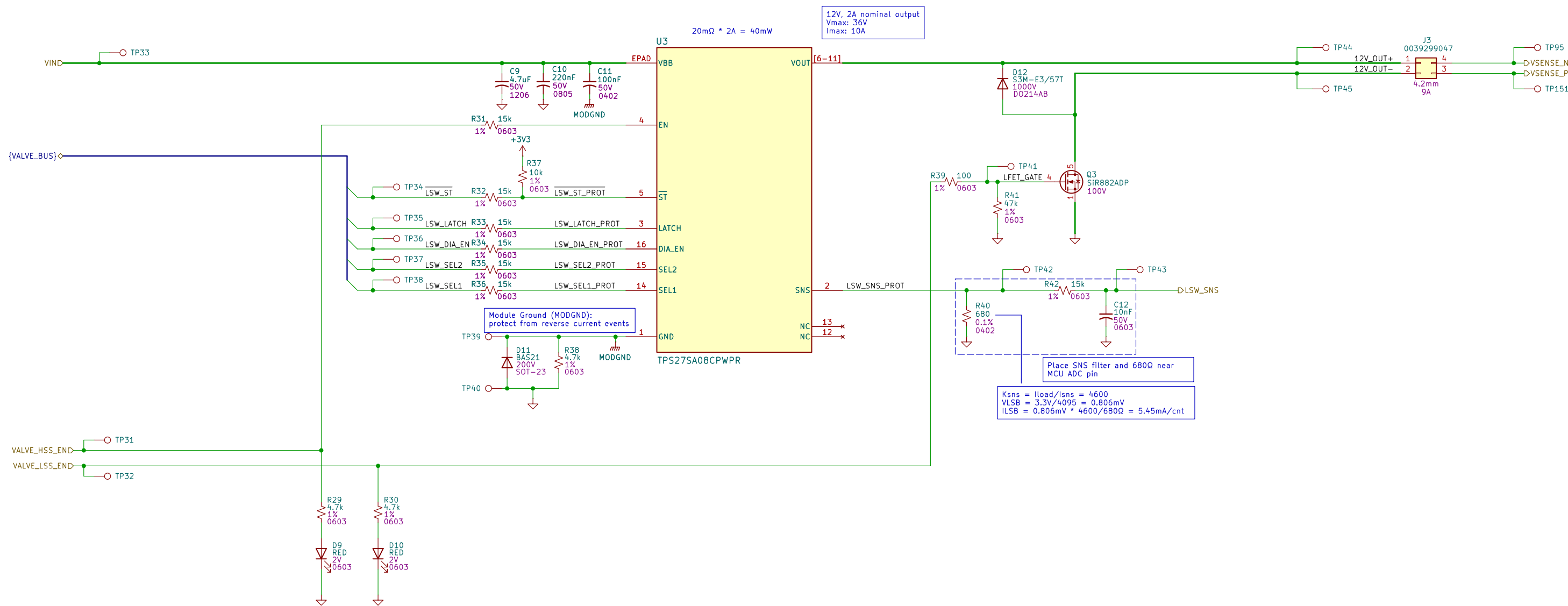
Id: 7/26

# VALVE OUTPUT POWER



Valve Enable Procedure:  
1. Enable common ELOAD (U26)  
2. Enable low-side switch (LSS)  
3. Enable high-side switch (HSS)

# VALVE OUTPUT POWER



# VALVE OUTPUT POWER



Id: 10/26

## VALVE OUTPUT POWER

[illegible]

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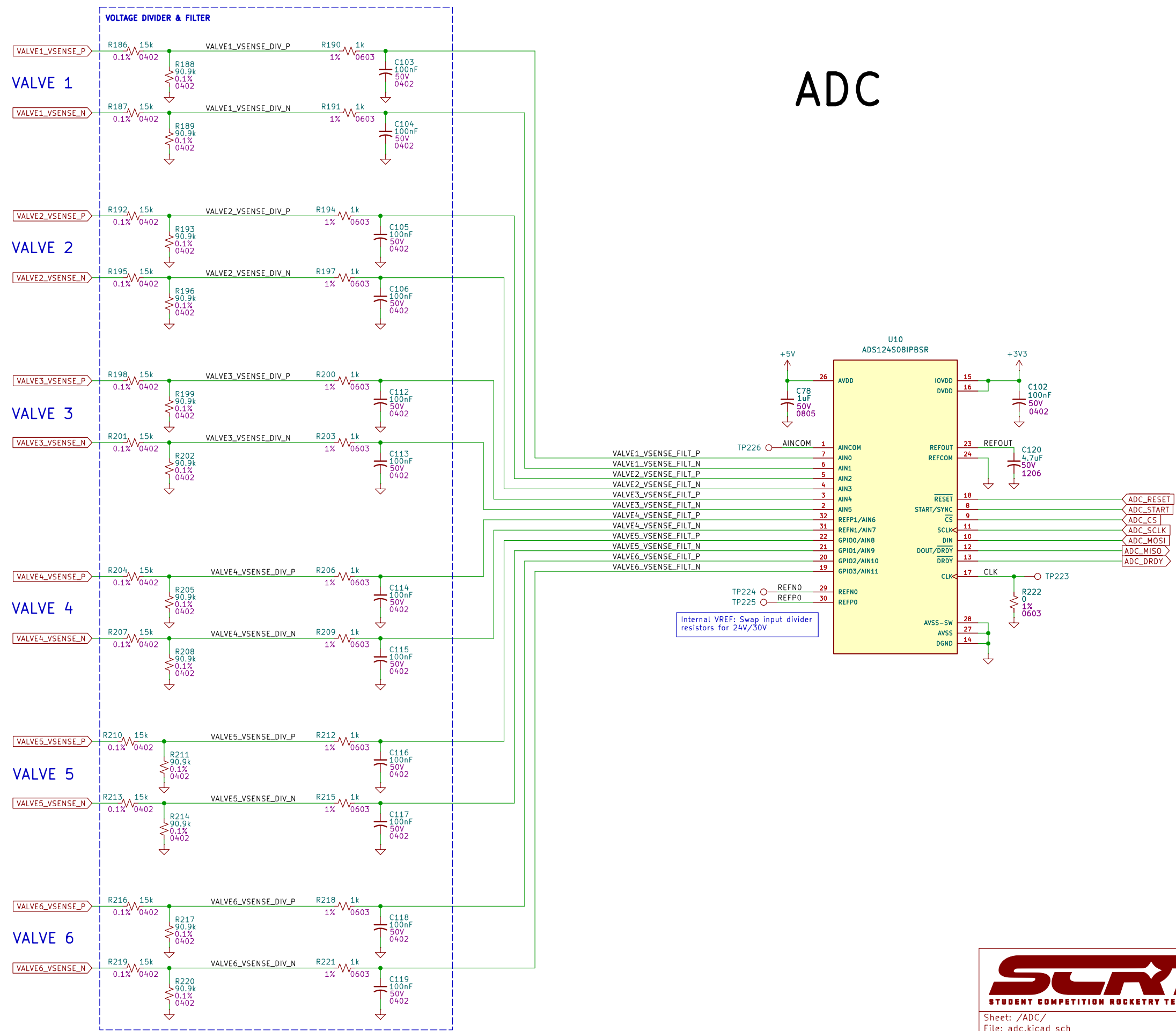
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# VALVE OUTPUT POWER

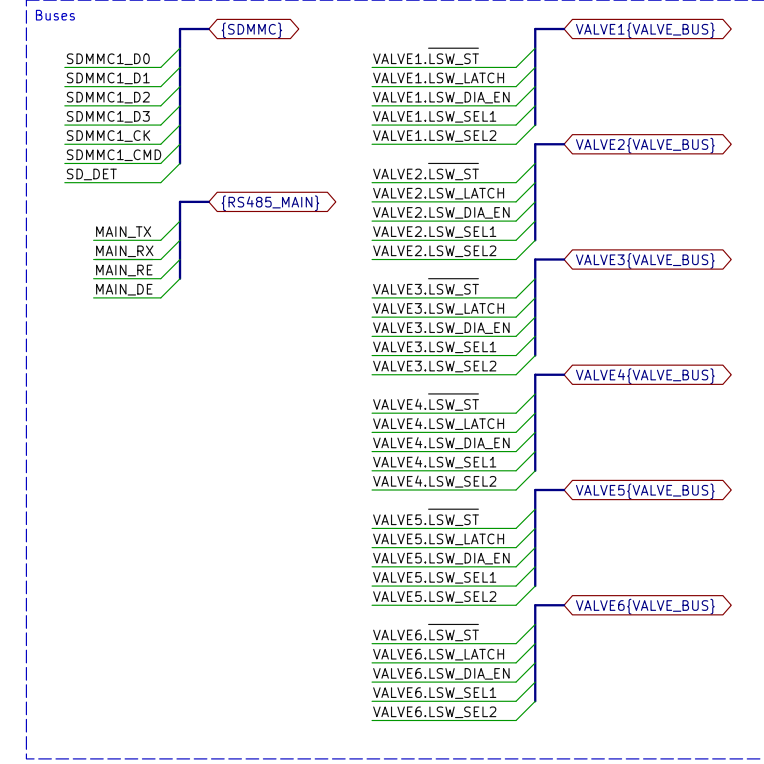
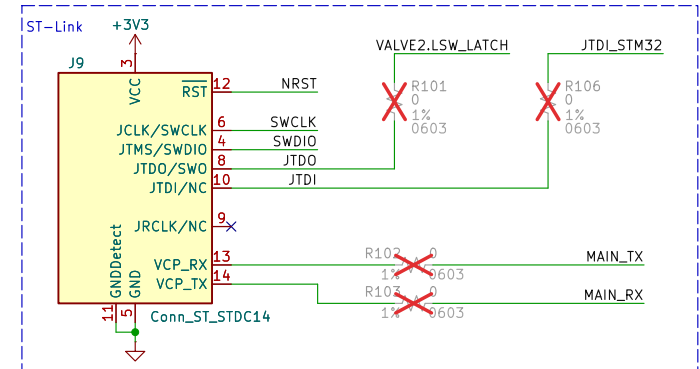
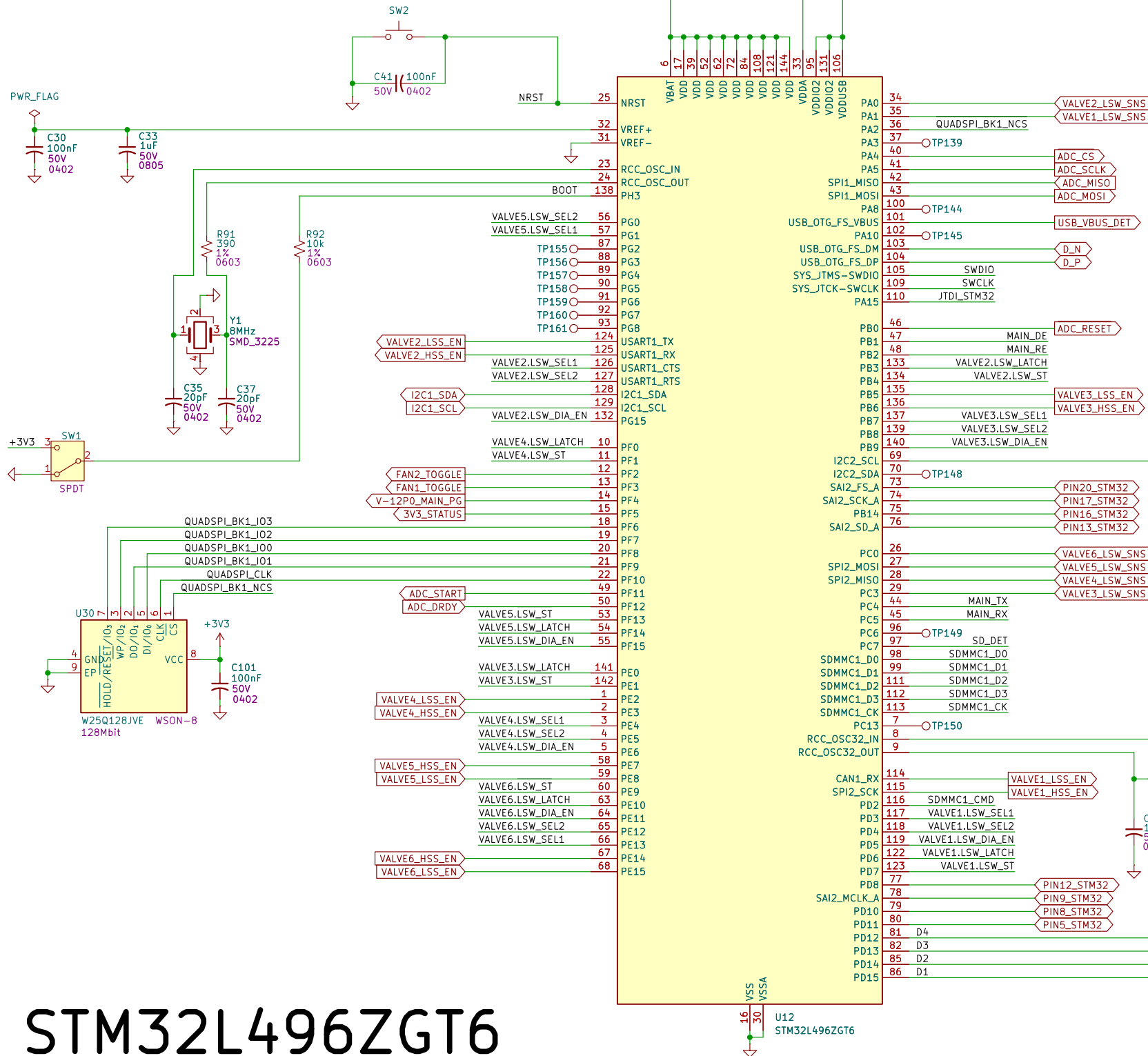
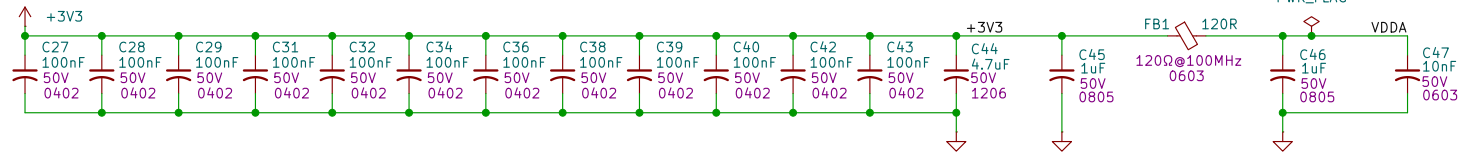


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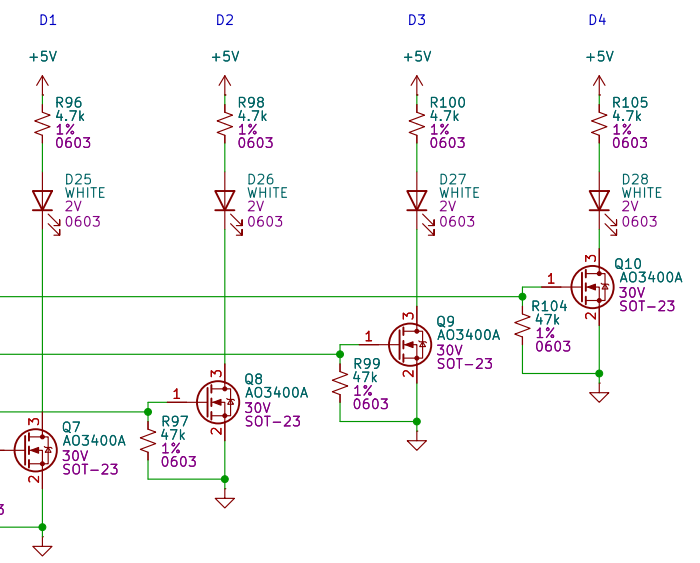
# ADC



# Supply & Analog Supply Filtering



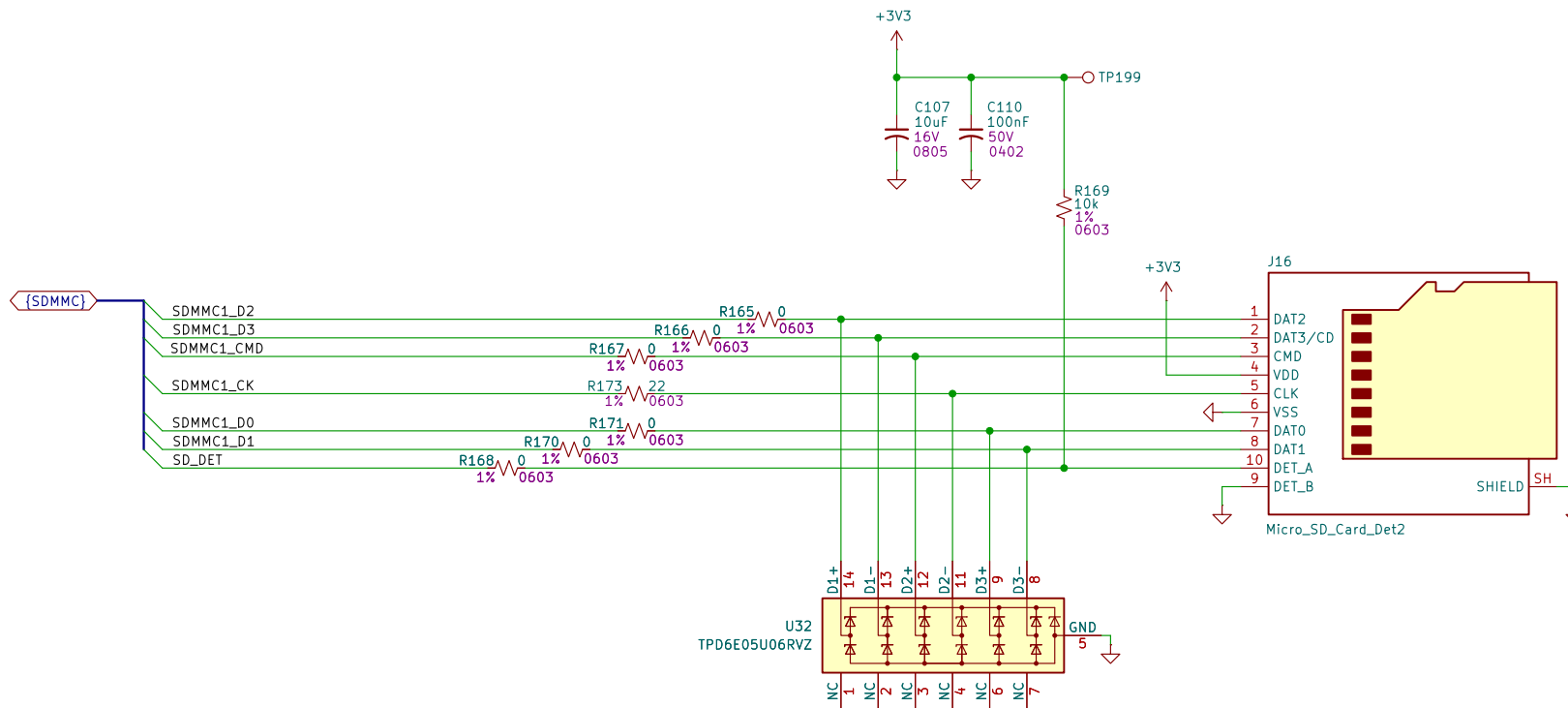
## Debugging and Status LEDs



STM32L496ZGT6



# SD CARD



Sheet: /SD Card/  
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**Title: SD CARD**

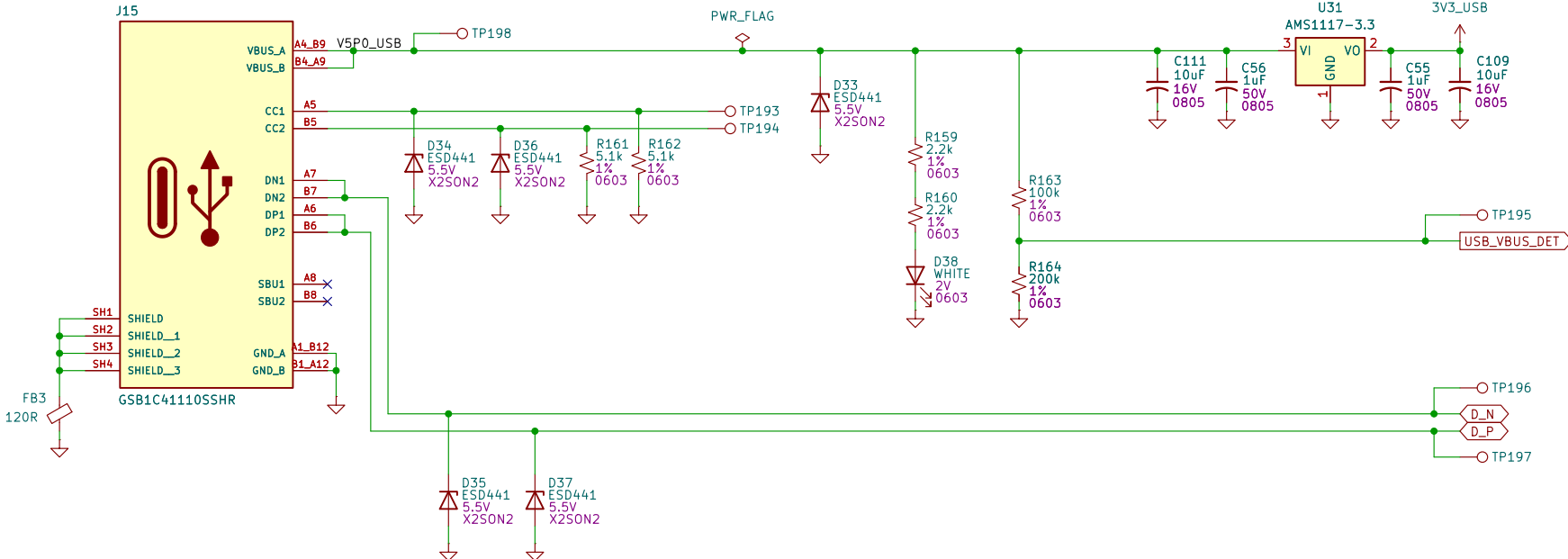
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| Size: A4 | Date: 2026-08-27 |
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# USB-C



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Size: A4

Date: 2026-08-27

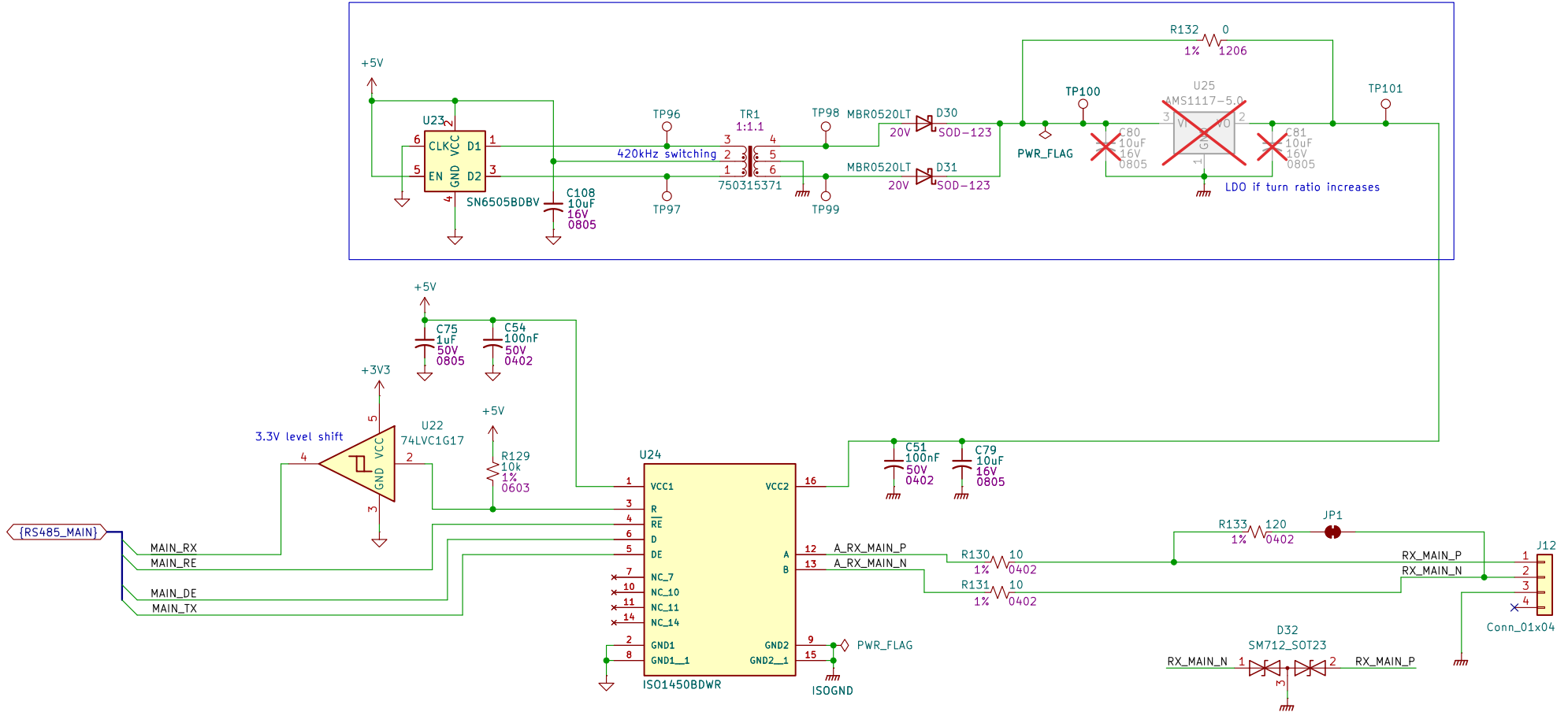
Rev: A

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Id: 16/26

# RS-485

## Galvanic Power Isolation



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File: rs-485.kicad\_sch

**Title: RS-485**

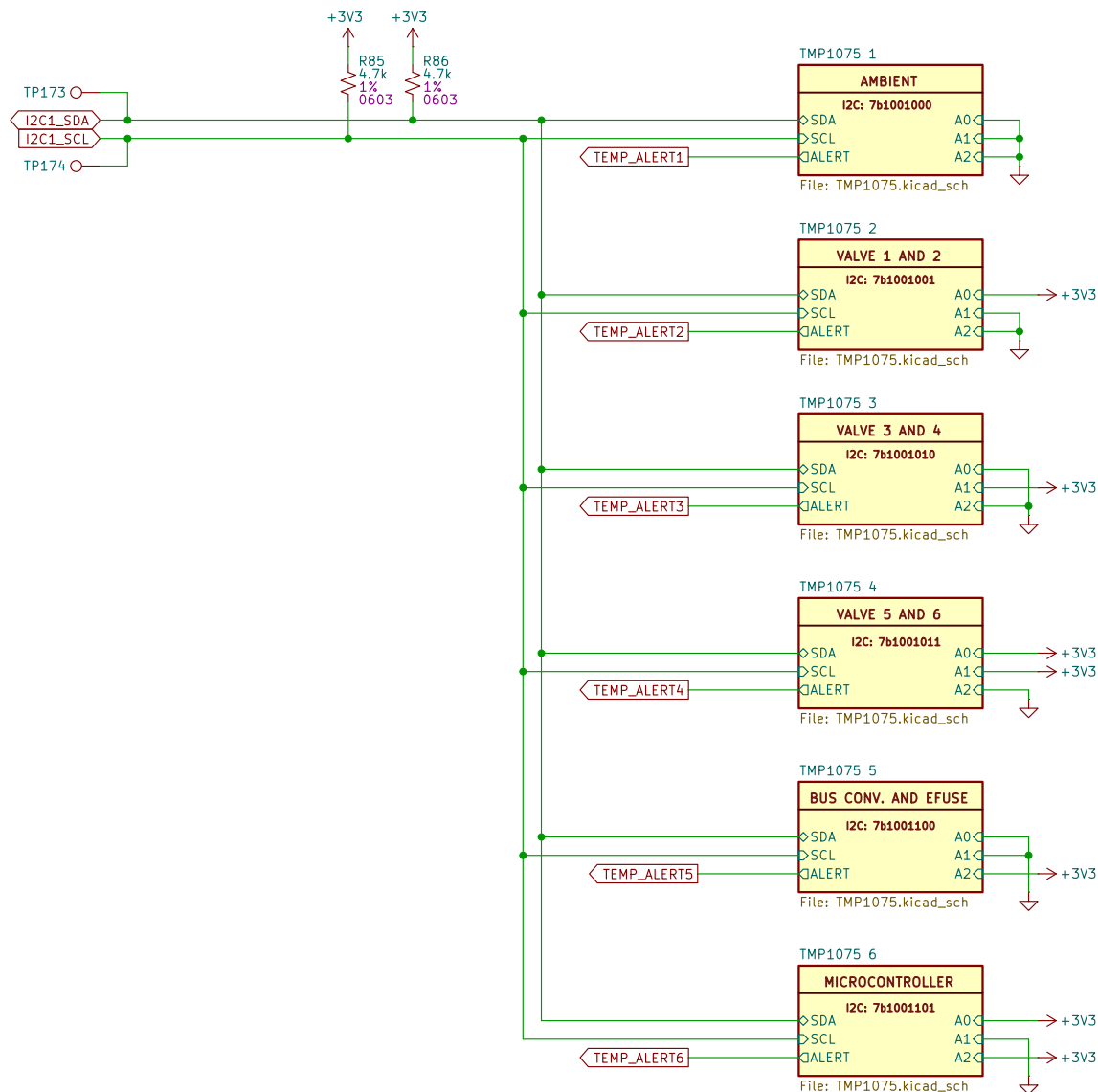
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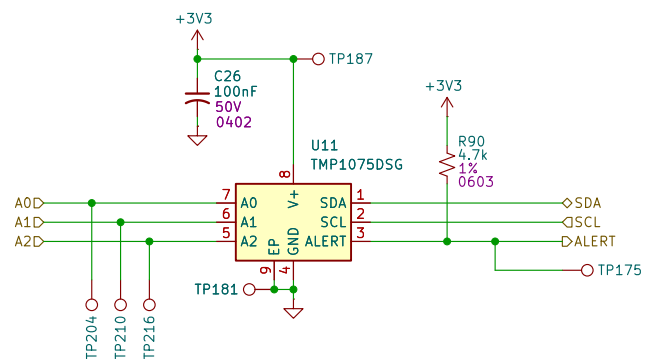
Rev: A

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# TEMPERATURE SENSORS



# TMP1075



Sheet: /Sensors/TMP1075 1/  
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**Title: TMP1075**

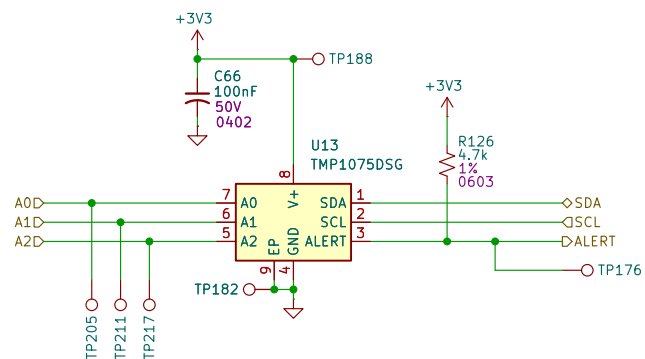
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# TMP1075



Sheet: /Sensors/TMP1075 2/  
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**Title: TMP1075**

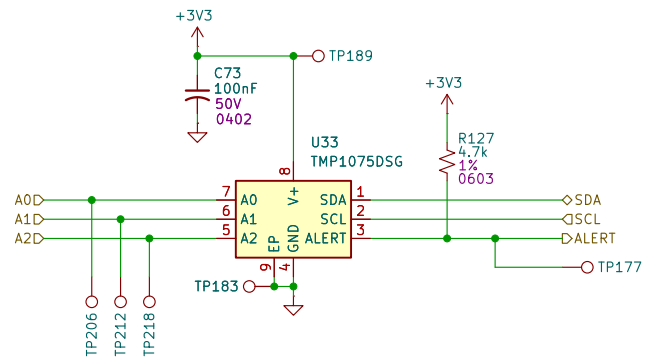
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# TMP1075



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**Title: TMP1075**

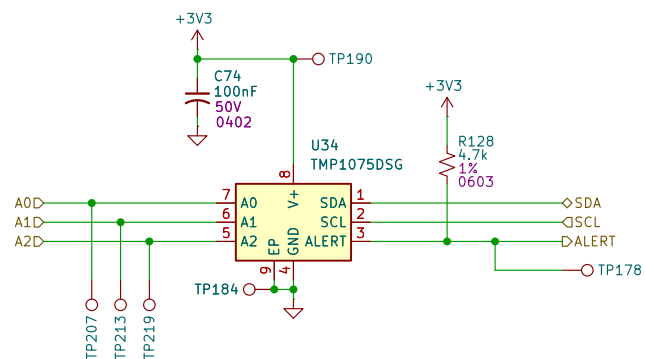
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# TMP1075



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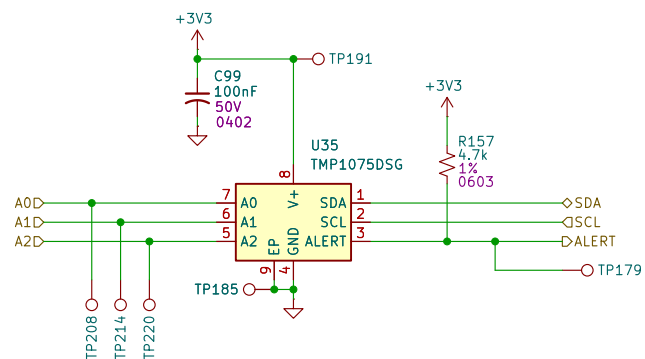
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# TMP1075



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**Title: TMP1075**

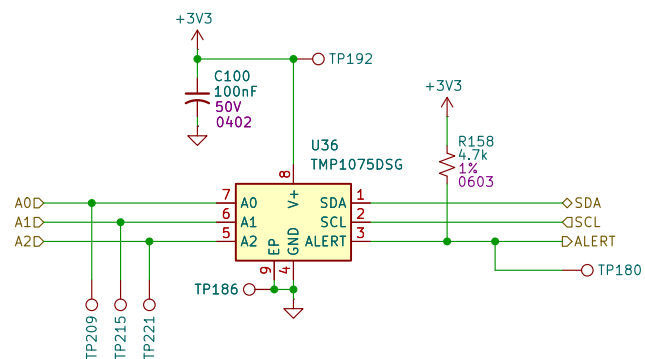
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# TMP1075



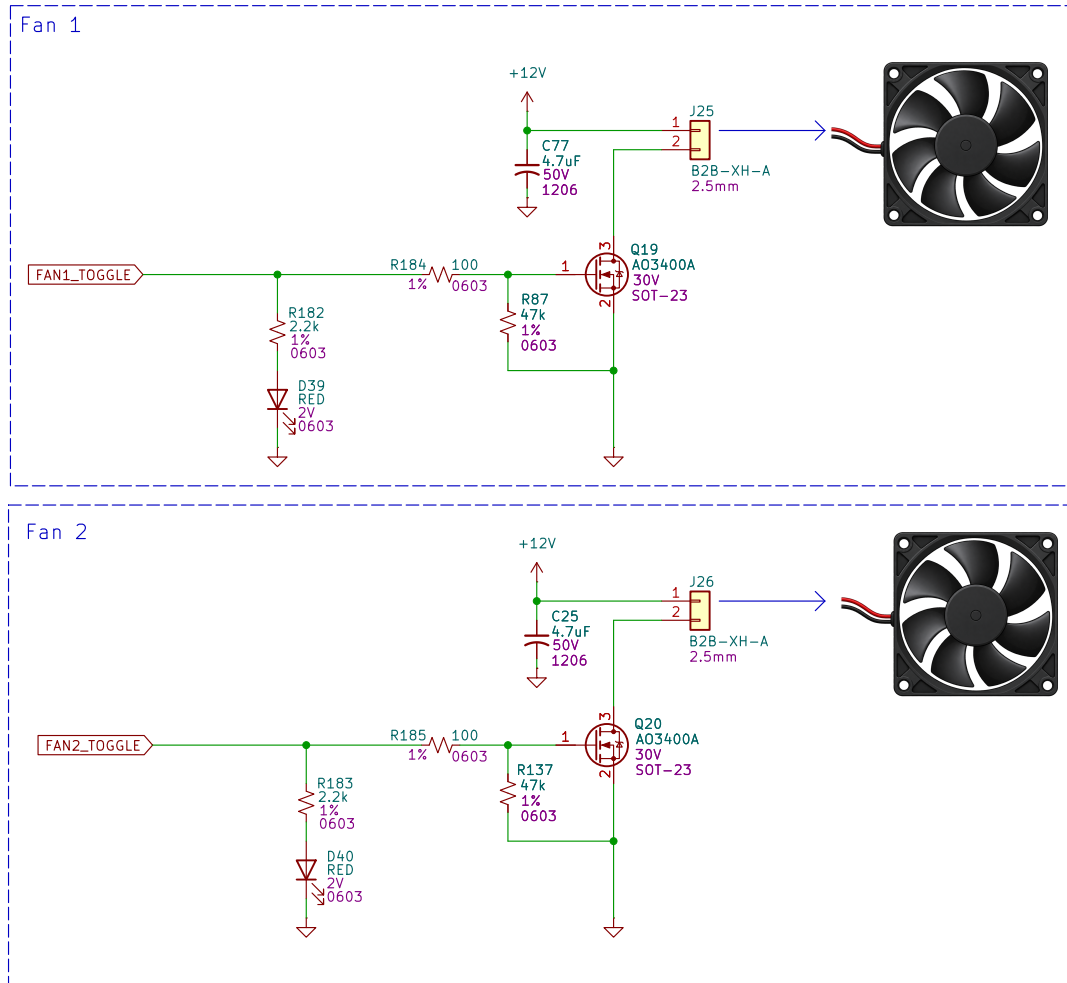
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# FANS



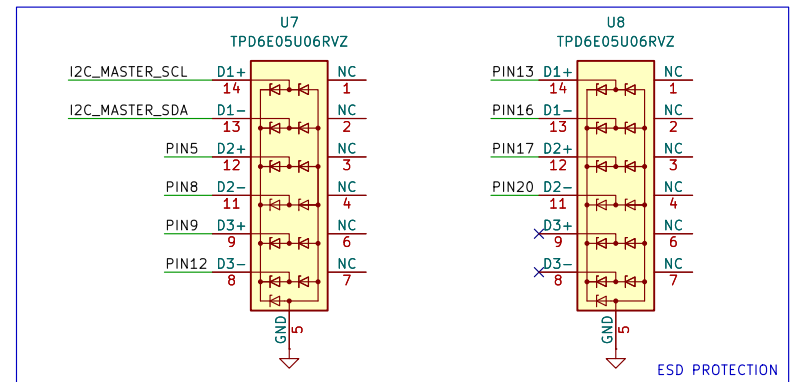
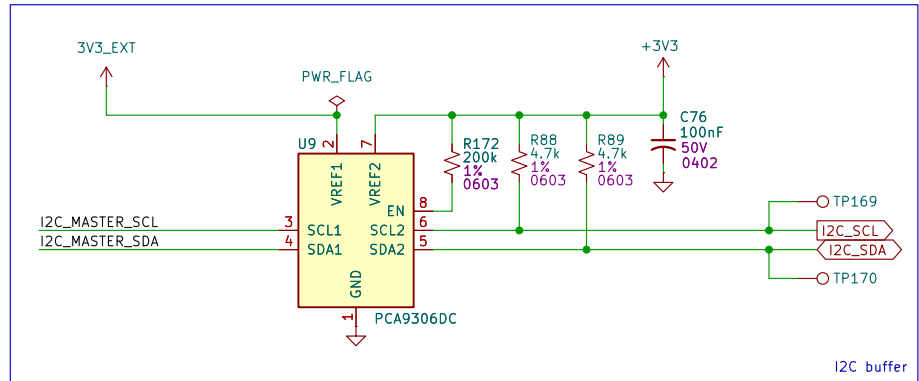
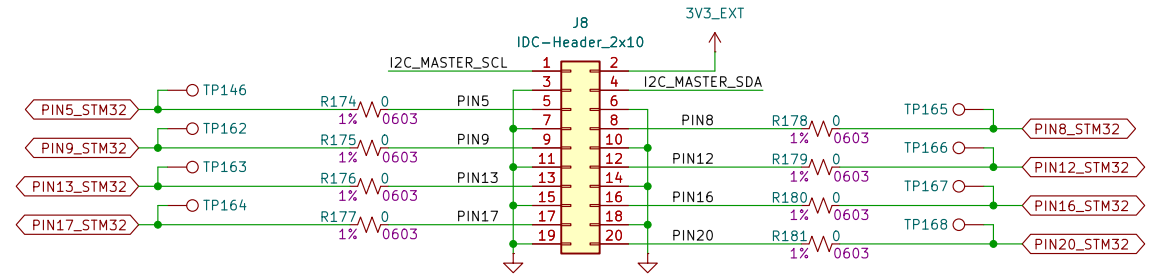
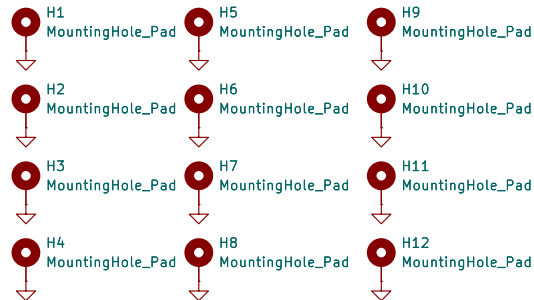
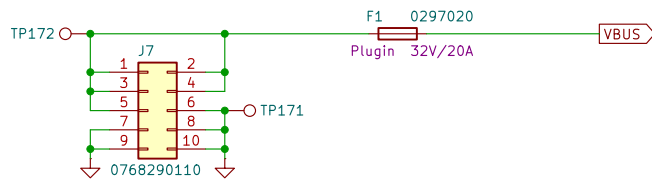
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# MECHANICAL



Sheet: /Connectors/  
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Size: A4 Date: 2026-08-27

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